

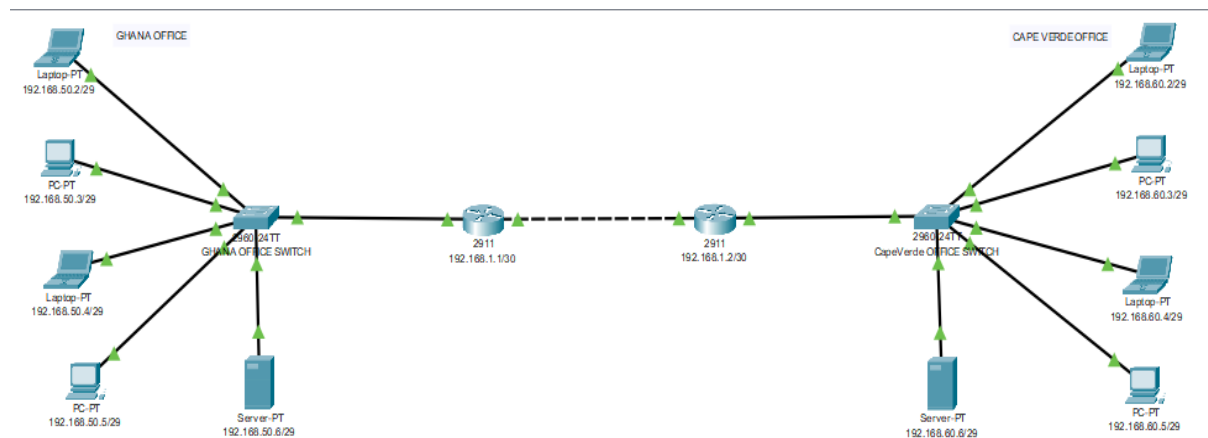
PROJECT: SECURE INTER-BRANCH CONNECTIVITY USING CISCO PACKET TRACER (DOCUMENTATION)

To design and simulate a secure WAN between two branch offices in Cisco Packet Tracer while allowing only HTTP and ICMP traffic, the following steps were implemented.

- ✓ Network topology
- ✓ IP addressing scheme
- ✓ Router configurations
- ✓ Access Control Lists (ACLs)
- ✓ Final verification steps

Network topology

A separate branch network was established for both branches (Ghana and Cape Verde); each branch comprises four end-user systems (a combination of PCs and laptops), one Switch, one Server, and one router, all connected via a WAN link.



A hybrid network topology with characteristics of both star and point-to-point topologies was chosen for this project. Here's a breakdown of the topology:

Hybrid Topology combining:

- Star Topology within each office (Ghana Office and Cape Verde Office)
- Point-to-point Link between the two routers connecting the offices

Ghana Office:

- Uses a star topology
- Devices (laptops, PCs, server) connect to a central switch (2960-24TT).
- The switch connects to a router (2911) with IP 192.168.1.1/30.

Cape Verde Office:

- Also uses a star topology
- Devices (laptops, PCs, server) connect to a central switch (2960-24TT).
- The switch connects to a router (2911) with IP 192.168.1.2/30.

Inter-office Connection:

- The two routers (192.168.1.1/30 and 192.168.1.2/30) are connected directly using a point-to-point link (copper cross-over cable).

This topology was chosen for the following reasons:

- A star topology within each office makes it easy to isolate and troubleshoot problems.
- If a single PC or cable fails, the rest of the network remains unaffected.
- The central switch acts as a hub for managing traffic efficiently.
- A point-to-point link between routers ensures a dedicated and direct connection between offices (Ghana ↔ Cape Verde).

IP addressing scheme

A private IP of GHANA: **192.168.50.0/24**, CAPE VERDE: **192.168.50.0/24**, and WAN LINK: **192.168.1.0/24**, was chosen and further subnetted to eliminate wasted addresses while meeting requirements and provide addressing for all branches.

Ghana Office Network:

- Network: 192.168.50.0/29
- Gateway: 192.168.50.1
- Subnet Mask: 255.255.255.248
- Usable Host Range: 6 usable IPs (192.168.50.1 - 192.168.50.6)
- Broadcast: 192.168.50.7

Cape Verde Office Network:

- Network: 192.168.60.0/29
- Gateway: 192.168.60.1
- Subnet Mask: 255.255.255.248
- Usable Host Range: 6 usable IPs (192.168.60.1 - 192.168.60.6)
- Broadcast: 192.168.50.7

WAN Link:

- Network: 192.168.1.0/30
- Subnet Mask: 255.255.255.252
- Usable Hosts: 2 usable Ips (192.168.1.1 (Ghana) and 192.168.1.2 (Cape Verde))
- Broadcast: 192.168.1.3

Router configurations

- Connect Ghana Router to Cape Verde Router using a Copper Crossover cable
- Connect Ghana Switch to Ghana Router using Copper Straight-Through cables
- Connect Cape Verde Switch to Cape Verde Router through Copper Straight-Through cables
- Connect all PCs and servers to their respective switches using Copper Straight-Through cables

```

Router>
Router>
Router>en
Router#configure t
Enter configuration commands, one per line. End with CNTL/Z.
Router(config)#hostname Ghana_Router
Ghana_Router(config)#interface gigabitethernet0/0
Ghana_Router(config-if)#ip address 192.168.50.1 255.255.255.248
Ghana_Router(config-if)#no shutdown

Ghana_Router(config-if)#
%LINK-5-CHANGED: Interface GigabitEthernet0/0, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/0, changed state to up

Ghana_Router(config-if)#interface gigabitethernet0/1
Ghana_Router(config-if)#ip address 192.168.1.1 255.255.255.252
Ghana_Router(config-if)#no shutdown

Ghana_Router(config-if)#
%LINK-5-CHANGED: Interface GigabitEthernet0/1, changed state to up

Ghana_Router(config-if)#ip route 192.168.60.0 255.255.255.248 192.168.1.2
Ghana_Router(config)#end
Ghana_Router#
%SYS-5-CONFIG_I: Configured from console by console

Ghana_Router#write memory
Building configuration...
[OK]
Ghana_Router#

```

Ghana Router configuration settings

End Device Configurations

Ghana Office:

- Laptop-PT: 192.168.50.2/29 - Gateway: 192.168.50.1, subnet: 255.255.255.248
- PC-PT: 192.168.50.3/29 - Gateway: 192.168.50.1, subnet: 255.255.255.248
- Laptop-PT: 192.168.50.4/29 - Gateway: 192.168.50.1, subnet: 255.255.255.248
- PC-PT: 192.168.50.5/29 - Gateway: 192.168.50.1, subnet: 255.255.255.248
- Server-PT: 192.168.50.6/29 - Gateway: 192.168.50.1, subnet: 255.255.255.248
- Ghana Router: 192.168.1.1/30 - subnet: 255.255.255.252

```

Router>
Router>
Router>en
Router#configure t
Enter configuration commands, one per line. End with CNTL/Z.
Router(config)#hostname CapeVerde_Router
CapeVerde_Router(config)#interface gigabitethernet0/0
CapeVerde_Router(config-if)#ip address 192.168.60.1 255.255.255.248
CapeVerde_Router(config-if)#no shutdown

CapeVerde_Router(config-if)#
%LINK-5-CHANGED: Interface GigabitEthernet0/0, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/0, changed state to up

CapeVerde_Router(config-if)#interface gigabitethernet0/1
CapeVerde_Router(config-if)#ip address 192.168.1.2 255.255.255.252
CapeVerde_Router(config-if)#no shutdown

CapeVerde_Router(config-if)#
%LINK-5-CHANGED: Interface GigabitEthernet0/1, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/1, changed state to up

CapeVerde_Router(config-if)#ip route 192.168.50.0 255.255.255.248 192.168.1.1
CapeVerde_Router(config)#end
CapeVerde_Router#
%SYS-5-CONFIG_I: Configured from console by console

CapeVerde_Router#write memory
Building configuration...
[OK]
CapeVerde_Router#

```

Cape Verde Router configuration settings

End Device Configurations

Cape Verde Office:

- Laptop-PT: 192.168.60.2/29 - Gateway: 192.168.60.1, subnet: 255.255.255.248
- PC-PT: 192.168.60.3/29 - Gateway: 192.168.60.1, subnet: 255.255.255.248
- Laptop-PT: 192.168.60.4/29 - Gateway: 192.168.60.1, subnet: 255.255.255.248
- PC-PT: 192.168.60.5/29 - Gateway: 192.168.60.1, subnet: 255.255.255.248
- Server-PT: 192.168.60.6/29 - Gateway: 192.168.60.1, subnet: 255.255.255.248
- Cap Verde Router: 192.168.1.2/30 - subnet: 255.255.255.252

Access Control Lists (ACLs)

```
Ghana_Router>
Ghana_Router>enable
Ghana_Router#configure terminal
Enter configuration commands, one per line. End with CNTL/Z.
Ghana_Router(config)#access-list 100 permit icmp any any
Ghana_Router(config)#access-list 100 permit tcp any any eq 80
Ghana_Router(config)#access-list 100 permit tcp any eq 80 any
Ghana_Router(config)#access-list 100 deny ip any any
Ghana_Router(config)#interface gigabitethernet0/1
Ghana_Router(config-if)#ip access-group 100 out
Ghana_Router(config-if)#
Ghana_Router(config-if)#show access-lists
^
% Invalid input detected at '^' marker.

Ghana_Router(config-if)#end
Ghana_Router#
%SYS-5-CONFIG_I: Configured from console by console

Ghana_Router#show access-list
Extended IP access list 100
 10 permit icmp any any
 20 permit tcp any any eq www
 30 permit tcp any eq www any
 40 deny ip any any
```

Ghana Access control list

```
CapeVerde_Router>
CapeVerde_Router>enable
CapeVerde_Router#configure terminal
Enter configuration commands, one per line. End with CNTL/Z.
CapeVerde_Router(config)#access-list 101 permit icmp any any
CapeVerde_Router(config)#access-list 101 permit tcp any any eq 80
CapeVerde_Router(config)#access-list 101 permit tcp any eq 80 any
CapeVerde_Router(config)#access-list 101 deny ip any any
CapeVerde_Router(config)#interface gigabitethernet0/1
CapeVerde_Router(config-if)#ip access-group 101 out
CapeVerde_Router(config-if)#end
CapeVerde_Router#
%SYS-5-CONFIG_I: Configured from console by console

CapeVerde_Router#show access-lists
Extended IP access list 101
 10 permit icmp any any
 20 permit tcp any any eq www
 30 permit tcp any eq www any
 40 deny ip any any
```

Cape Verde Access control list

The Access Control Lists (ACLs) are configured to:

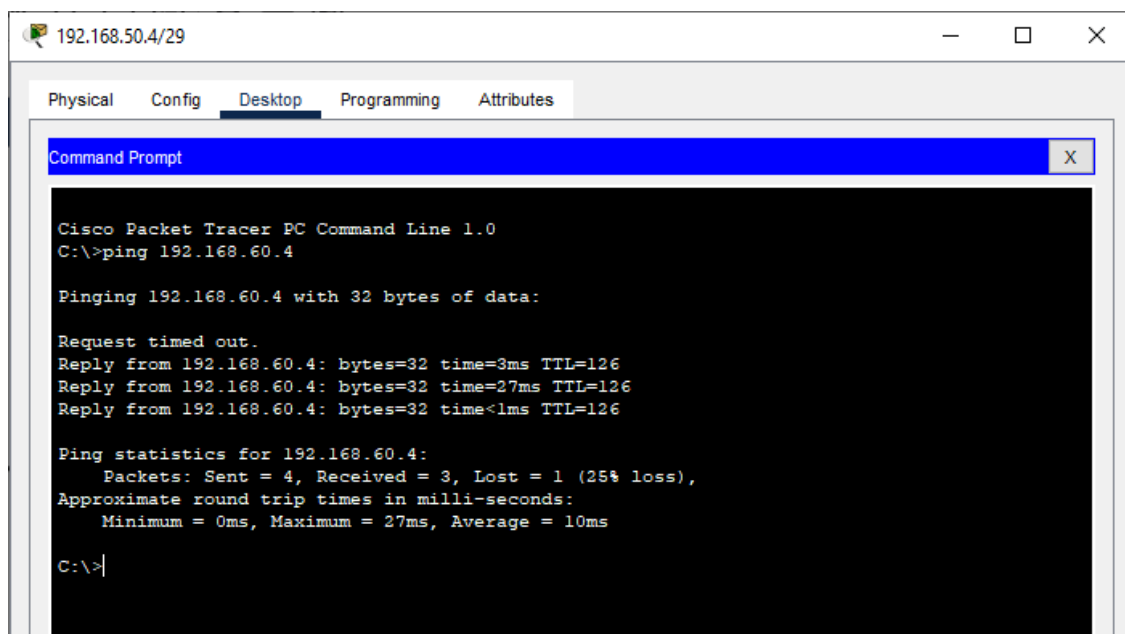
- ✓ Permit HTTP traffic (TCP port 80) in both directions between the networks
 - Permit tcp any any eq 80: allows traffic going to web servers (http requests) from any source IP to any destination IP using port 80
 - Permit tcp any eq 80 any: allows traffic coming from web servers (http responses) from any source IP using port 80 to any destination IP

- ✓ Permit ICMP traffic (ping) in both directions between the networks
 - Permit icmp any any: allows ping between branches from any source IP to any destination IP
- ✓ Explicitly deny all other IP traffic
 - Deny ip any any: blocks all other protocols and ports from any source IP to any Destination IP

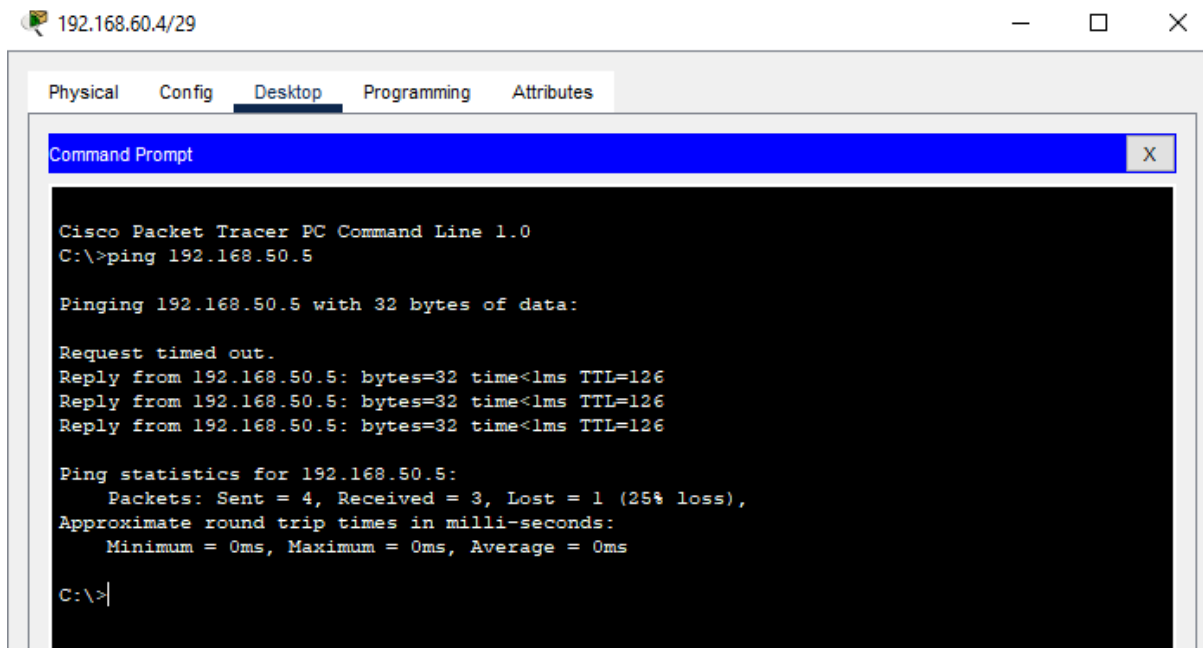
The ACLs are applied outbound on the Copper Crossover interfaces of both routers to filter traffic leaving the router towards the other branch.

Final verification steps

1. ICMP Testing:

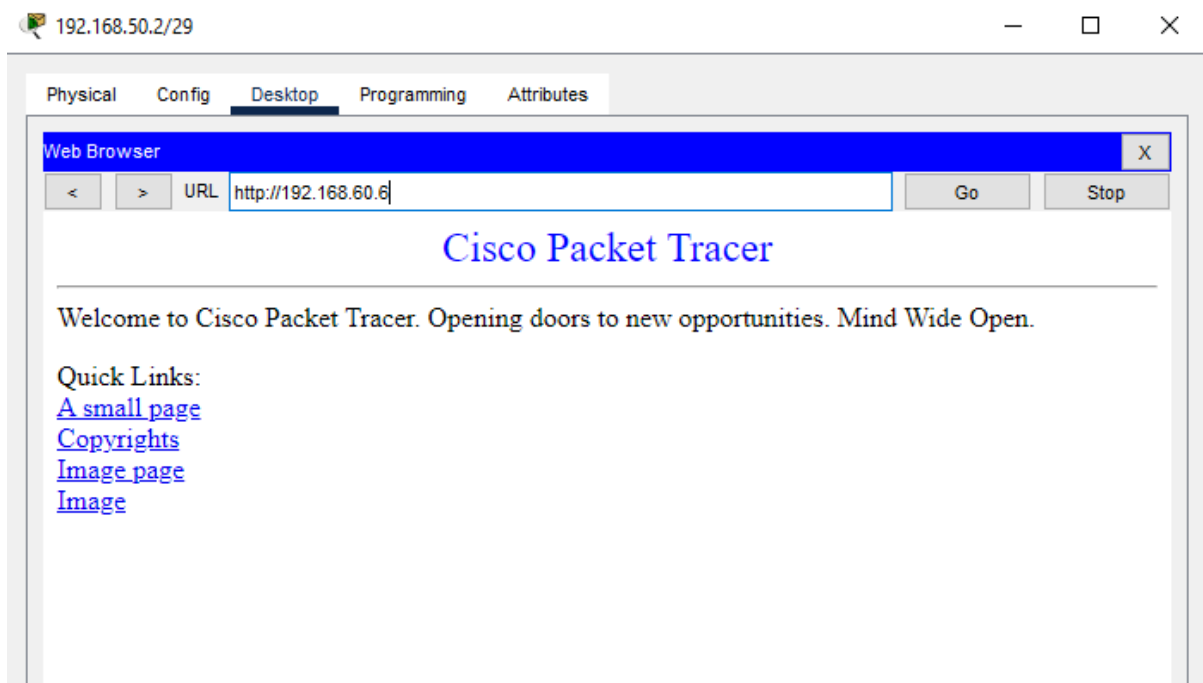


From Laptop-PT (192.168.50.4) in Ghana, a ping request was carried out on 192.168.60.4 (Laptop-PT in Cape Verde), which was successful as shown above.

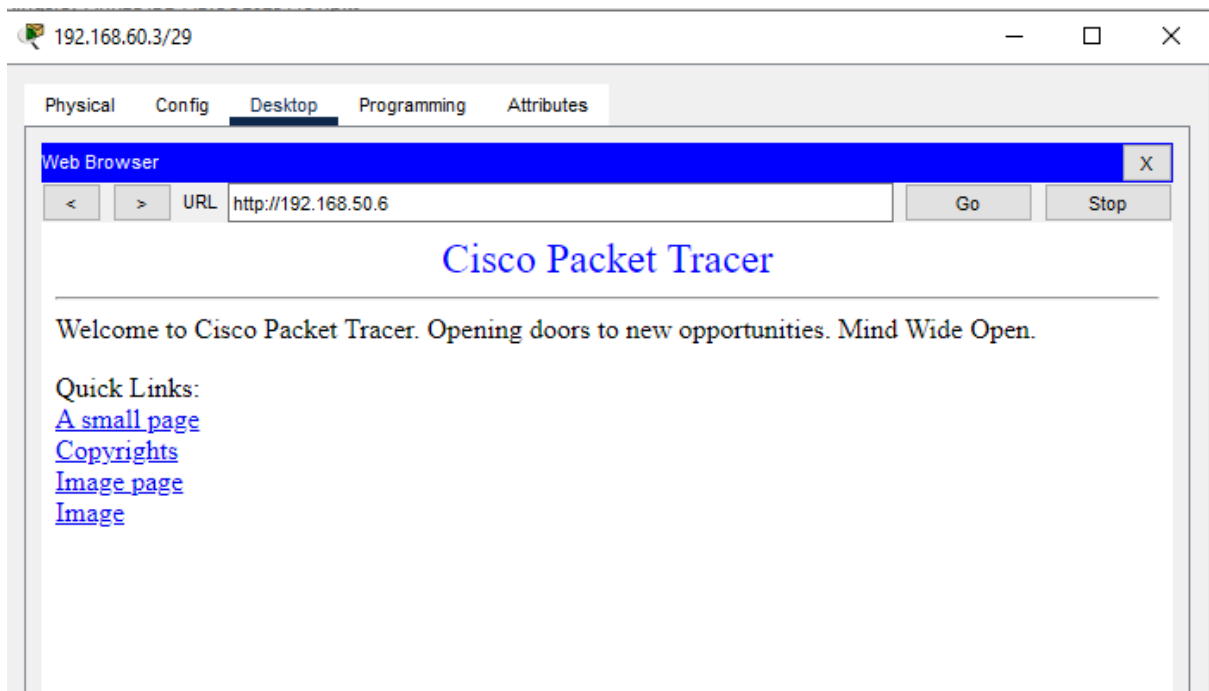


From Laptop-PT (192.168.60.4) in Cape Verde, a ping request was carried out on 192.168.50.5 (PC-PT in Ghana), which was successful as shown above.

2. HTTP Testing:

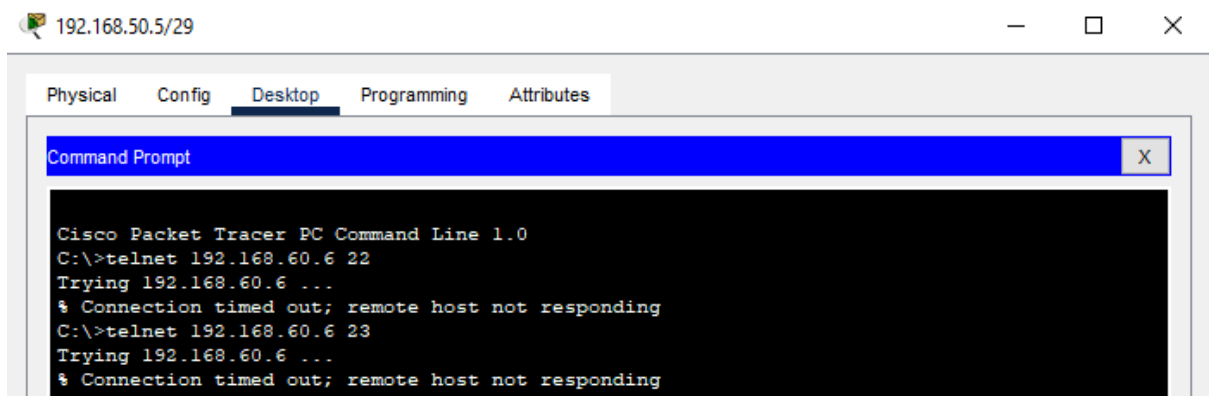


From Laptop-PT (192.168.50.2) in Ghana, open a web browser to http://192.168.60.6 (Cape Verde server), which will successfully connect as shown above.

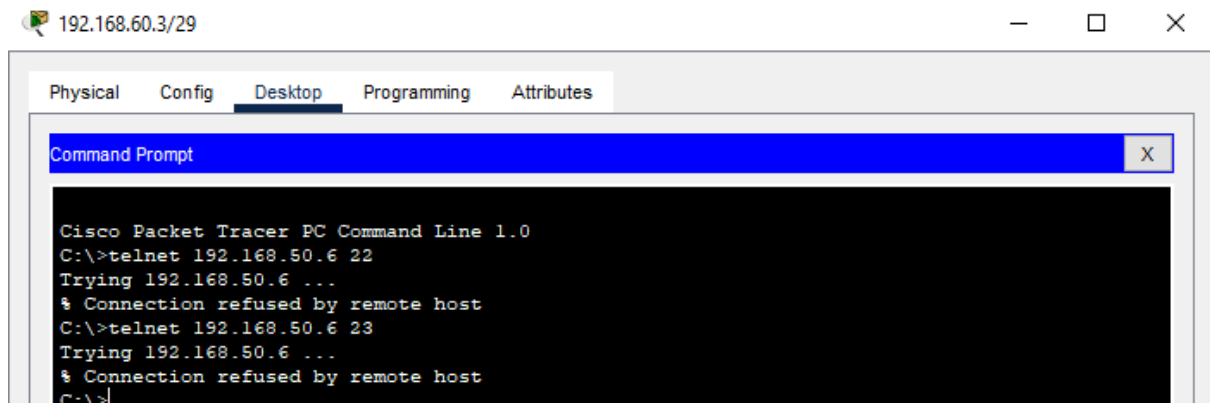


From PC-PT (192.168.60.3) in Cape Verde, open a web browser to <http://192.168.50.6> (Ghana server), which will successfully connect as shown above.

3. Blocked Traffic Testing:



Attempt to SSH and Telnet from PC-PT (192.168.50.5) to 192.168.60.6 in the Cape Verde branch failed



Attempt to SSH and Telnet from PC-PT (192.168.60.3) to 192.168.50.6 in the Ghana branch failed